

Practice Exercises

1 Filesystem Hierarchy Standard (FHS)

- List the main directories under / and explain their purposes.
- Navigate to /var/log and list its contents.
- Use ls -l to check the file type and permissions of /etc/passwd.

2 Finding Files and Commands

- Use locate to search for bashrc. Why might the results be outdated?
- Update the file database and run locate again.
- Search for all .conf files in /etc using find.
- Find all files modified in the last 3 days.

3 Using find for Advanced Searches

- Find all files larger than 5MB in /var/log.
- Locate empty files in /home.
- Search for directories under /usr that do not belong to any user.
- Run find with -exec to list details of found files like ls -l.

4 Command Location & Execution

- Use whereis to locate the ls binary and its documentation.
- Use which to determine the exact path of grep.
- Find out if multiple versions of passwd exist on the system.

Week 7 complete lab workflow

Part A – Create test files

```
mkdir -p ~/week7-lab/subdir  
cd ~/week7-lab
```

```
touch report.txt  
touch emptyfile  
touch subdir/settings.conf  
touch subdir/application.log
```

Create a file larger than 5 MB:

```
dd if=/dev/zero of=largefile.bin bs=1M count=6
```

Verify:

```
ls -lh  
find . -maxdepth 2
```

Part B – Test **locate**

Update the database:

```
sudo updatedb
```

Search:

```
locate week7-lab  
locate report.txt
```

Create another file:

```
touch newfile  
locate newfile
```

It might not appear.

Update again:

```
sudo updatedb  
locate newfile
```

It should now appear.

Part C – Search by name and type

Find `.conf` files:

```
find ~/week7-lab -type f -name "*.conf"
```

Find directories:

```
find ~/week7-lab -type d
```

Find empty files:

```
find ~/week7-lab -type f -empty
```

Expected empty files include:

```
report.txt  
emptyfile  
newfile  
settings.conf  
application.log
```

unless content has been added to them.

Part D – Search by size and time

Find the file larger than 5 MB:

```
find ~/week7-lab -type f -size +5M
```

Expected:

```
/home/student/week7-lab/largefile.bin
```

Find files modified in the last ten minutes:

```
find ~/week7-lab -type f -mmin -10
```

Part E – Use **-exec**

Display details of all files:

```
find ~/week7-lab -type f -exec ls -lh {} \;
```

Display only **.conf** file details:

```
find ~/week7-lab -type f -name "*.conf" -exec ls -l {} \;
```

Part F – Locate commands

```
whereis ls  
whereis passwd  
which grep  
which passwd  
which -a passwd
```

Part G – Clean up

```
rm -rf ~/week7-lab  
sudo updatedb
```

Common Week 7 mistakes

Searching with **locate** immediately after creating a file

```
touch testfile
```

locate testfile

The file may not appear until:

sudo updatedb

Forgetting the starting directory in **find**

Better:

```
find /etc -name "*.conf"
```

The starting directory determines where the search begins.

Not quoting wildcards

Risky:

```
find /etc -name *.conf
```

Correct:

```
find /etc -name "*.conf"
```

The quotation marks ensure that **find** receives the pattern.

Confusing **-mtime -3** and **-mtime +3**

Expression	Meaning
-mtime -3	Newer than three days
-mtime +3	Older than three days

Using **-empty** without specifying a type

```
find /home -empty
```

This returns both empty files and empty directories.

For empty files only:

```
find /home -type f -empty
```

Forgetting the end of `-exec`

Incorrect:

```
find /etc -name "*.conf" -exec ls -l {}
```

Correct:

```
find /etc -name "*.conf" -exec ls -l {} \;
```

Searching the whole system without hiding errors

```
find /
```

may produce many permission errors.

Use:

```
find / 2>/dev/null
```

Week 7 essential commands

Command	Purpose
<code>locate PATTERN</code>	Search the file-location database
<code>locate -i PATTERN</code>	Case-insensitive database search
<code>updatedb</code>	Update the <code>locate</code> database
<code>find PATH -name PATTERN</code>	Search by case-sensitive name
<code>find PATH -iname PATTERN</code>	Search by case-insensitive name
<code>find PATH -type f</code>	Find regular files

<code>find PATH -type d</code>	Find directories
<code>find PATH -mtime -3</code>	Find recently modified files
<code>find PATH -mmin -10</code>	Find files modified recently
<code>find PATH -size +5M</code>	Find files larger than 5 MB
<code>find PATH -user USER</code>	Find files owned by a user
<code>find PATH -nouser</code>	Find entries with no valid owner
<code>find PATH -empty</code>	Find empty files or directories
<code>find ... -exec</code>	Run a command on each result
<code>whereis COMMAND</code>	Find binaries and documentation
<code>which COMMAND</code>	Show executable selected from <code>PATH</code>
<code>which -a COMMAND</code>	Show all matching executables in <code>PATH</code>